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Bridging the Gap Between Stewards and Creators

Clashes between bottom line–oriented managers (stewards) and creative technical employees (creators) may be inevitable. But when those conflicts aren't managed well, a company's ability to innovate may be at risk.

Robert D. Austin and Richard L. Nolan

Most technology-intensive companies today depend on specialized and talented employees, in fields ranging from high-tech product development to life sciences research. Such employees often design much of their own work; no one else is as qualified as they are to do so. The work itself may involve intangible materials and products. Often managers can't tell what exactly is going on. As one CEO said, frustrated by his inability to manage a technical staff, "Sometimes they're hanging out drinking coffee, other times they're rushing around — I don't know what any of it means."

Never before in history have there been such profound knowledge gaps between managers and the frontline employees who create business value. In earlier times, supervisors often rose from the ranks of workers and therefore were expert in the activities being supervised. But even when that is true today — when a software developer becomes a development manager, or a genetics researcher becomes a research manager — the specialized field in which value creation occurs keeps moving forward. A software developer more than four or five years removed from actual coding is no longer expert. With the exception of a few remarkable individuals, most managers can't keep up with all the details of advancing technology while also having full-time management responsibilities.

Being a good supervisor traditionally meant encouraging sound business practices and introducing changes to those practices as business conditions changed. Lately, something has gone radically wrong in the second part of that formulation, at least for businesses in fast-changing industries that rely on specialized employees. Now the changes sometimes come from key employees whose work managers don't completely understand. This wouldn't be a significant problem if managers and specialized employees always saw eye to eye. But they don't. The worldviews of these two groups can differ drastically. Consequently, the process of business innovation can resemble a series of battles between people with very different priorities. Such conflict often entails a substantial risk of squandering corporate resources.

The idea of an ongoing struggle between results-oriented managers and technical visionaries is not new. Economist and sociologist Thorstein Veblen noted it in his 1904 book *The Theory of Business Enterprise*.[1] Eighty-some years later, John Kenneth Galbraith cited Veblen's view to

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describe a dynamic still at work in a more modern economy:

"The businessmen, for good or ill, keep the talents and tendencies of the scientists and engineers under control and suppress them as necessary in order to maintain prices and maximize profits. From this view of the business firm, in turn, comes an obvious conclusion: somehow release those who are technically and imaginatively proficient from the restraints imposed by the business system and there will be unprecedented productivity and wealth in the economy."[2]

The difficulties managers can face when supervising employees with specialized skills have not gone unnoticed. In the late 1960s, management consultant Peter Drucker pointed out both the business importance of highly specialized "knowledge workers" and the challenges of managing them.[3] Psychologists have established that more creative outcomes result when creators are intrinsically rather than extrinsically motivated, a fact that sets the stage for conflict between managers trying to improve the bottom line and employees whose job responsibilities include creative work.[4] Moreover, motivational techniques such as piece rates and bonus schemes, which work in traditional employment settings, have been shown to work badly when products are novel or intangible, as is the case in many technologically specialized activities.[5] George Mason School of Public Policy professor Richard Florida has suggested that the special characteristics of

About the Research

This research employed a qualitative, inductive approach aimed at discovering principles of innovation used in development of the Internet, which is often cited as a major human achievement.[i] Although the adjacent fields of innovation and psychology of creativity research are relatively mature, we considered our methodology a good fit with our research objectives, because we were interested primarily in a historically exceptional instance of the innovation process (the development of the Internet), rather than more common practices and patterns. Consistent with a sound inductive approach, we closely examined surrounding literatures in order to identify areas of consistency and difference.

We began by reviewing the large literature documenting the history of computing and the Internet, including books, research papers, popular media and Internet archives. Many histories are quite detailed, and we were able to discern people and organizations frequently mentioned in numerous accounts. Using various sources, we were also able to identify members of groups who were not mentioned as often but who were involved in the same efforts as frequently mentioned players. We thus completed lists of group membership as much as possible. In total, we identified more than 60 pioneers involved in the origins of the Internet and modern computing.

We then mapped individual pioneers into groups termed "houses." Houses corresponded in some cases to physical locations (for example, Xerox PARC or MIT Lincoln Laboratory), but in other cases they were "virtual" (for example, the ARPA Internet community, physically located at many universities). By carefully tracking the movements of pioneers among houses between about 1962 and the mid-1990s, we discerned waves of "migration" that could be set in historical context. Three migrations are of particular significance: (1) migration from ARPA as Congress inserted stipulations into defense funding authorizations; (2) migration to Xerox as the corporate lab was set up, staffed and gained notoriety in the early and mid-1970s; and (3) migration from Xerox PARC as corporate headquarters increased control and as PARC's projects were passed over in choosing Xerox's next generation of office automation products in the late 1970s.

The two principal researchers and a research assistant worked independently and together to fill in gaps in the patterns of movement and to reconcile inconsistencies that appear in the literature.

This inductive approach caused us to become interested in the movements of particular people who seemed to be involved in a disproportionate number of Internet-related developments. Even people who did not come up in the popular stories (who might not have been as good at self-promotion) tended to reappear in "houses" that generated significant accomplishments.

To obtain some interpretation of the logic behind these events, we interviewed about a dozen pioneers, including Vint Cerf, Douglas Engelbart, Alan Kay, Bob Taylor and Dave Walden. These subjects were chosen partly because of their centrality in our maps and in historical accounts, but also because we could gain access to them. Open-ended interviews prompted the interview subjects to recount chronologies and interpretations of events, in order to identify causal factors to explain the movements we had seen in our maps.

These interviews brought into stark relief the conflict between the people we have labeled stewards and creators. Through analysis of the interviews by three different people (the two primary researchers and a research assistant), we developed the notions of stewards and creators. We further enhanced the development of these ideas by drawing on our related research on innovation and our field research in companies such as Cisco Systems, Ford, Novell and Sun Microsystems.

i. In 2005, for example, CNN listed the Internet as the No. 1 nonmedical invention since 1980, according to a panel of technology leaders assembled by the Lemelson-MIT Program. See "Top 25: Innovations," June 19, 2005, www.cnn.com/2005/TECH/01/03/cnn25.top25.innovations.

workers in the "creative class" require adjustments to many of our ideas about management and business development.[6]

Indeed, in a study we conducted on innovation in the development of the Internet, we found evidence of disputes and misunderstandings between the people whom we call *stewards* (usually managers) and others we call *creators* (usually specialized, highly capable employees) which can cause delays in business adoption of new technologies. Stewards care most about allocating resources efficiently and responsibly. Creators, in contrast, care most for the glittering vision and higher purpose, and they often see business concerns as secondary. The two perspectives naturally conflict, and disputes inevitably arise. Both stewards and creators contribute vitally to business innovation. Thus, when the stewards and creators within an organization are too much at odds with one another, or when the balance of power between the two groups is too skewed, the potential harm is much greater than mere squandered resources. In such situations, a company's ability to innovate effectively may be at risk.

Our conclusions about stewards and creators and their roles in innovation stem from a research project that entailed identifying more than 60 people deeply involved in the origins of the Internet and modern computing — people who could be safely designated as Internet pioneers. (See "About the Research.") We then tracked how those Internet pioneers migrated among various organizations over a period of more than three decades. We also interviewed some of the pioneers in order to better understand the movements of these technical experts from one organization to another. Through analysis of those interviews, we developed the concepts of stewards and creators.

The Long Wait For a Commercial Internet

Most businesspeople tend to think of the Internet as a phenomenon that happened rapidly. The commercial Web is what they have in mind. Web commerce began in earnest in August 1995 with Netscape Communications Corp.'s initial public offering, expanded furiously until March 2000 and then abated somewhat, leaving a substantial infrastructure in place. A great deal happened in a short period of time. But some computing pioneers have a very different perspective, such as Bob Taylor, who headed the Internet initiative at the U.S. Department of Defense's Advanced Research Projects Agency in the late 1960s and then led a group at Xerox Corp.'s Palo Alto Research Center in the early 1970s. He observed: "People tell me the Internet happened very fast. They're crazy. It took forever!"

A quick review of the facts reveals that Taylor has a point. Thanks to Taylor and others, the first four nodes of the Internet were installed by December 1969. Taylor and his colleagues tried to alert the business community to the potential of the new technology, but

their efforts were rebuffed. In Taylor's view, the 26-year interval between the birth of the Internet and Netscape's IPO, when businesses finally seemed to see value in the technology, was too long.

The reasons Taylor's entreaties were not better received are no doubt complex. But the potential was there, in the long-term vision of certain creators, for more rapid innovation. Should it have taken more than 25 years for business to wake up to the promise of the Internet? And do similar delays happen even now in the day-to-day process of business innovation?

When we interviewed computing pioneers — people like Taylor, Alan Kay (a principal inventor of the personal computer, object-oriented programming and graphical user interfaces) and Douglas Engelbart (who, along with his team at Stanford Research Institute, invented the mouse as well as technologies that were important predecessors to the now-familiar icon-based desktop computer interface) we expected to find them content, pleased with the extent of their influence. Instead, they expressed impatience, even bitter disappointment over missed opportunities. The current commercial Internet, remarkable as it is, falls short of what these creators thought they'd see by now. While it is debatable whether their expectations are reasonable or even desirable from a societal point of view, their feelings show the gap between creator and steward mindsets. At Xerox PARC in the 1970s, for example, managers and technology visionaries "communicated like creatures of different species."[7]

Understanding Stewards and Creators

The contrast between the mindsets of stewards and creators is substantial. (See "Stewards vs. Creators.") However, the dichotomy is conceptual; it would be hard to find pure examples of either mindset in real individuals. But in discussions of the steward/creator concept with innovators, managers and researchers, these two labels resonate powerfully. Better understanding of these two contrasting roles might be a key to shortening the time required to bring innovations to market.

Creators want to realize a grand vision, usually technical but often including social implications. They frequently have a deep connection between commitment to a mission that extends beyond daily work concerns and the intellectual talents and intrinsic motivation that make top-notch workers so valuable. In other words, talented creators' commitment to what they do is a large part of the reason they are so good at their work. Engelbart, for example, expressed his overriding objectives and his attitude toward them by saying, "What if I can maximize the value of my career and help humankind? … You have to have a crusade." He described his specific vision as "augmenting human capabilities," that is, using technology to equip humankind better for the challenges of an increasingly complex world. Similarly, artificial intelligence pioneer J.C.R. Licklider and Bob Taylor had a grand vision, expressed in their 1968 paper "The Computer as a Communication Device,"[8] for how people would use computers for "creative, interactive communication."

The governing impulse for stewards, in contrast, can be summarized by the phrase "the law of diminishing returns." Stewards don't want to allocate one dollar past the point where it achieves greater recompense for the firm. Bound by notions of professional responsibility for optimizing the application of assets to produce a return, stewards seek not grand visions but products, services and business solutions good enough to maximize the economic benefits the organization can capture. To stewards, products that are too good from a technical point of view — better than can justify the investment required to develop them — are as much a problem for the business as products that aren't good enough.

Creators don't understand stewards' willingness to settle for good enough. Their commitment to the grand vision makes creators unwilling to compromise, and they often interpret compromise as failure. For example, Engelbart's efforts to augment human capabilities led him to invent not just the mouse but also other computer interface devices that took time to learn to use but extended human resourcefulness. His five-chord keyset, for example, when used with a mouse, allowed users to interact with a computer in a

Stewards vs. Creators

The process of technological innovation within a company often includes clashes between stewards, who are often managers, and creators, who are generally highly skilled technical employees.

	Stewards	Creators
Governing Impulse	Allocate investor resources responsibly	Realize the grand vision at all costs
Attitude on Point of Diminishing Return	Going beyond it would be bad business	Don't know where it is, don't care where it is
Attitude on Compromise	An inevitable part of many practical business solutions	Not to be considered when trying to achieve a vision
Question Always in Mind	What's the business model?	Is it "cool," fun, exciting or potentially important to humankind?
Orientation in Producing Economic Value	Value capture	Value creation

Kay points to the restraint that stewards often exercise over creators and offers a hypothesis that seems to complement Veblen's: "Business would rather control mediocre people than be out of control with talented people."

unique and productive manner without a full keyboard. However, business thinkers endorsed a different principle: user-friendliness. That was a practical decision aimed at gaining quick acceptance for new products. When the idea of user-friendliness caught on, it sidelined any device that took time to learn to use. Engelbart saw lost opportunity in that outcome. With his National Medal of Technology and a picture of him shaking hands with the president of the United States both hanging on the wall above him, Engelbart expressed disappointment: "I feel like a failure … The world didn't seem to change."

Creators frequently zero in on stewards' goals and attitudes when pointing out reasons for their thwarted visions. They often disdain businesspeople with steward mindsets and instead want to spend time with other talented creators. Said Taylor: "For 35 years, I never worked with anyone who I didn't want to work with. … The people I did work with were really good. Lots of IQ points. They weren't just good, they were really good." And Alan Kay said: "The world is run by C-pluses. ARPA [in the late 1960s when Taylor was there] aimed at A-plus-plus; no one asked the A-plus-pluses what they were going to do."

Kay's comment points to the restraint that stewards often exercise over creators. Talented creator employees who think of themselves as contributors to human progress have ideas about what they should do that may not map perfectly to a business plan. Such employees may drift outside business goals. They don't allocate effort to suit managers. Because these workers are so smart — often orders of magnitude more productive than possible replacements — partial alignment with management goals may produce excellent business outcomes. Even so, stewards often exert control over creators that creators resent. Engelbart said of the "profit-oriented businessmen" he knew: "People like this shot me down again and again. They said 'You don't understand the market.'" Kay offered a hypothesis that seems to complement Veblen's: "Business would rather control mediocre people than be out of control with talented people."

The Internet pioneers often cited efforts to control the productive but difficult to manage technical visionaries as a reason that periods of creative ferment came to an end. Taylor and others suggested that the burst of inventive activity that produced the Internet ended when the U.S. Congress legislated that all funds awarded by ARPA had to go to demonstrably defense-related projects. As sensible as this may seem, in practice the legislation resulted in new control regimes and stopped people like Taylor from granting funds to researchers based on the "quality of their nervous systems."[9] In the aftermath of this legislation, Taylor and others grew disillusioned and left their ARPA-funded posts and projects.

Taylor landed at PARC, where he recruited many people who he knew were extremely talented. The exceptionally creative period there lasted until the Xerox senior management in Connecticut clamped down on their freewheeling West Coast researchers. One notorious episode that may have signaled the beginning of the end involved publication of a feature article about PARC in *Rolling Stone*, which corporate managers back in Connecticut regarded as a "druggie magazine." "Corporate executives could only conclude from their insular perch in Stamford that PARC was reeling out of control, shamelessly squandering the research facility's budget on adolescent techno-fantasy trips rather than solid, marketable scientific pursuits."[10] When corporate managers exerted increased control over PARC, many creative employees fled. Taylor himself departed for Digital Equipment Corp., a computer company that had its headquarters in Maynard, Massachusetts.

We mapped influential Internet and computing pioneers by the assorted communities and organizations they were part of. The pattern revealed in the migration of Internet and computing pioneers among various organizations reflected two factors: (1) reaction against stewardlike control structures and (2) a desire to be part of communities of like-minded creators. Creators' resentment of control might be discounted if they were never right. But they sometimes are. By now, the ideas and inventions of Engelbart, Kay, Taylor and other Internet pioneers have led to tens, if not hundreds, of billions of dollars of value created and captured. And although it's possible to argue that conditions weren't right for capturing the value of these ideas and inventions when they originally arose, it might also be true that conditions *were* right. For example, some pioneers like Bob Metcalfe, who was originally at Xerox PARC, left established companies and founded new ones that created and captured value that could have belonged to their former employers. (Metcalfe left PARC and founded 3Com Corp. in 1979.)

In reality, managers' inclination to restrain creator tendencies rather than pay attention to them sometimes may pose a danger to a business. Managers may know best when it comes to packaging products or services for sale, distributing them and selling them; concerns about whether value is "harvestable" might legitimately lead managers to pass on an investment that excites creators. But

companies can't harvest what they won't plant or grow.

On the other hand, creators aren't always right, either. Some activities of creators — such as a star programmer who pursues a pet technical project while the company is facing a critical, looming deadline that needs his full attention — are surely counterproductive. It's also not hard to find cases of companies that chased innovation for its own sake and were not successful in the business arena; many Internet start-ups during the Internet boom of the late 1990s fit in that category.[11]

Managing the Steward-Creator Relationship

The struggle between stewards and creators often gets dangerously out of balance. In some cases, risk-averse product marketers may block the plans of a development organization trying to bring unproven technologies to market. As one leader in a major technology company's laboratories complained in the late 1990s: "We keep coming up with 190 miles-per-hour fastballs that no one in the company wants to catch." Alternatively, technology gurus may push products into the market even though there is no clear business model underlying them. Such behavior was rampant during the Internet bubble. When forces are too unequal in either direction and either stewards or creators get too much of an upper hand, innovation breaks down. Maintaining a healthy balance between stewards and creators provides the best path to business innovation.

Keeping a good balance requires a deep understanding of how the perspectives of stewards and creators differ. Most managers who oversee innovation processes find themselves inclined toward the steward's point of view. In organizations that emphasize earnings per share and short-term results, some managers feel intolerant or downright antagonistic toward creators' obliviousness to business realities. Creators, meanwhile, do not hesitate to reciprocate such antagonism.

Stewards and creators engage in disputes both as peers and in supervisor/subordinate relationships. The "go/no-go" meeting for a new product or service is a classic battlefield. It ultimately falls to senior managers of the innovation process to mediate disputes and coax valuable results out of the conflict. Senior innovation managers must control their affinity for the viewpoint of either steward or creator, restrain the counterproductive inclinations of their testy subordinates and bring out the best traits of both.

Is conflict and misunderstanding between creators and stewards inevitable? Yes, to some extent. But that doesn't mean it's not possible to manage the innovation process more effectively with management techniques that may minimize both waste and lost opportunity. Some companies do successfully bridge the gap between steward and creator mindsets. From a previous, broader study of innovation,[12] we have derived eight guidelines for reducing the potentially destructive impact of steward-creator conflict.

1. Keep creators around. This point may seem obvious, but not all companies do a good job of retaining talented creator employees, for two reasons. First, the vexation stewards feel with creator attitudes can sometimes prompt rash actions that leave a company with too few or insufficiently talented creators in key areas. An executive who succumbs to the urge to fire a difficult-to-manage but talented developer is an example of this problem. Second, stewards sometimes make adjustments in culture or environment that seem sensible or innocuous from their point of view but that antagonize creators. And when creators begin to leave, the process can accelerate due to their inclination to work with like-minded people. A CEO might intend to fire just one hard-to-control developer, but two or three other highly talented employees might depart as well. Putting up with a certain amount of maddening behavior by creators may be a price worth paying to keep great talent.

2. Balance influence between stewards and creators. Creators often feel unappreciated by stewards and impatient or even bitter when stewards do not appreciate the importance of a greater vision. As Harvard Business School professor Clay Christensen has argued,[13] many incumbent firms fail to exploit disruptive technologies that do not help sustain today's business but have the potential for surpassing existing technologies. The earliest advocates of disruptive technologies within a firm are usually creators who see, or sometimes merely sense, potential in a new technology, although it may not be the business potential that most interests them. Because their sense of a new technology is experienced in terms of seeing it as "cool" or exciting, creators are sometimes not very good at making the case for a disruptive technology in terms that are attractive to stewards.

Managers of innovation can at least partially address these challenges by following a simple rule: *Don't let the stewards always win*. When creators are arguing a case and stewards don't see a viable business model, keep the project alive to gain more information about the thing creators feel so strongly about. Doing this sometimes will go a long way toward keeping creators happy, or less unhappy, and it may occasionally generate an innovation blockbuster. Of course, managers should also follow the opposite rule: *Don't let the creators always win*. However, instances when creators gain the upper hand may be less common and shorter-lived in most established organizations.

3. Cultivate bridging personalities. Some people have credibility in both steward and creator camps. Such people can generate and capture economic value without being pure stewards, and they can appreciate a mysterious new technology without being pure creators. In early stages, the new thing that excites creators often appears formless to stewards; bridging individuals, though they are not creators, see the emerging form sooner and can begin to

Some people have credibility in both camps. They can capture economic value without being pure stewards; they can appreciate a mysterious new technology without being pure creators. Take Bill Gates, for example.

evaluate its potential. Such individuals are rare but extremely helpful; they often are capable of reducing difficulties by mediating disagreements and explaining points of view that conflict.

Bob Taylor, although he had more in common with creators than stewards, often acted as a bridging person. He described this role, in part, as helping identify and resolve what he called "Class 1" and "Class 2" disagreements. A Class 1 disagreement, according to Taylor, was one in which "the parties who disagree cannot describe to each other's satisfaction the other party's point of view." A Class 2 disagreement, on the other hand, was one in which "each party can describe completely to the other party's satisfaction the other party's point of view." Taylor maintained that parties could work effectively on problems involving Class 2 disagreements even if they could not agree, but not on problems involving Class 1 disagreements. Bridging personalities, by virtue of their credibility to both parties and their ability to understand and explain points of views in each camp, can help convert Class 1 conflicts into Class 2 conflicts. This valuable skill can resolve steward-creator conflicts or problems between creators.

Other well-known people in the history of the computer industry have operated as effective bridging personalities. Most people would say that the true genius of Microsoft Corp.'s Bill Gates is as a businessperson, but he also has the know-how to talk with engineers about technology. Eric Schmidt, now CEO of Google Inc., acted as a bridging personality during an important episode when he was CEO of Novell Inc.; Novell's engineers had resisted adopting open standards for the company's networking products, but Schmidt's technical credibility allowed him to move smoothly between the worlds of stewards and creators to produce agreement.[14]

The positive effect bridging personalities have on a company's performance stems from their ability to reduce uncertainty about potential returns from investments in research and development. Bridging personalities help disagreeing creators work out their differences, which helps stewards come to a better understanding of possibilities. Often, bridging personalities can reinterpret the positions of creators in a way that allows stewards to achieve more effective business judgments.

4. Use peer review to provide more accurate evaluation. Because work today has become specialized, standard forms of supervisory evaluation are not very effective in creative work; they merely antagonize talented creators. A time-tested approach to managing such situations exists, however: peer review. At PARC, Taylor routinely required his colleagues to review one another's work. Given widely varying reports, he could reduce considerably the uncertainty attending the innovative work of his group of technical superstars. Peer review has difficulties of its own, such as problems with ideological bias. Nevertheless, it is another way of bridging knowledge asymmetry between managers and highly capable workers

5. Structure the innovation process to regularly produce tangible artifacts. Structuring work during the innovation process so that creators produce tangible artifacts at regular intervals provides another control checkpoint for managers and another opportunity to reduce outcome uncertainty. At PARC, Taylor routinely asked employees to "describe their projects" to each other. Cisco Systems Inc., while engaging in the complex, specialized work of enterprise resource planning system installation, used "rapid iterative prototyping" to arrive at detailed descriptions of how the system would, say, process an order or check the availability of an item in inventory.[15] Budget commitments can be linked to the production of artifacts, allowing an additional level of control.

Artifacts need to be good tools for discussing future possibilities. One thing that does not qualify for this purpose is the traditional milestone document. Required milestone documents too often become busywork for employees and are unintelligible. A high-quality artifact produced as part of an innovation process should evoke a clear view of future possibilities and facilitate a detailed, realistic conversation. The artifact may be a document, but thought must go into what constitutes a useful one.

6. Realize that there will always be some conflict. The objective in managing highly capable employees is not to eliminate all conflict. Taylor's project description sessions and his peer evaluations were contentious, as was hiring at PARC. Some conflict can be a good thing: Different opinions can be useful for determining where the truth really lies. However, overly emotional, untrustworthy responses have the opposite effect. Harvard Business School professor Dorothy Leonard has called the tendency toward useful conflict "creative abrasion"; she observes that it keeps teams pushing for better solutions and prevents unconstructive uniformity from setting in.[16] The creative caldron needs to bubble, but it should not boil over.

7. Avoid overly prescriptive control mechanisms. When legislation began to specify in the late 1960s that ARPA funding had to be applied to demonstrably defense-related purposes, this by itself did not constitute a prescriptively detailed control mechanism. But when bureaucrats instituted controls that would allow them to prove that they were complying with the new legislation, they doomed the creative ferment around ARPA. Taylor could no longer award grants based on a past history of doing interesting work. Creative researchers lost the flexibility they needed to explore widely enough to stay interested. Researchers happy to comply with the detailed requirements in order to get funding weren't as creative as those that fled. PARC followed a similar trajectory; if the office had been located in Connecticut under the thumb of Xerox headquarters staff, instead of in California, the exceptionally creative outburst that occurred there in the mid-1970s might never have happened.

8. Manage the rate of convergence on closure. The steward instinct judges the progress of a project by its proximity to closure. The rush to finish sooner, to be efficient, to stay on or get ahead of schedule often predisposes stewards to make and encourage decisions earlier rather than later. But hurrying to decisions may hinder the innovation process. Rather than forcing decisions as early as possible in a creative process, managers may do better to make sure closure is achieved neither too fast nor too slow. Projects that converge on closure too quickly foreclose options and preclude high-value outcomes unnecessarily. Creators hate seeing this happen and will fight it, often tooth and nail. Projects that converge too slowly, however, are not ready to produce results when they are needed. Managers can help keep closure from coming too quickly by siding with creators to prevent stewards from forcing too-early decisions. Alternatively, sometimes managers have to push in the other direction, toward closure, by encouraging agreement on a point under dispute or, as a last resort, making the decision themselves.

An Awkward Partnership

Managers with steward instincts who follow the advice laid out here should reduce the uncertainty they encounter in predicting the outcomes of various possible research or product-development investments. This is the most important reason to tame steward reflexes and make allowances for the inexplicable behavior of some creators. Manage creators poorly, and a company can miss huge opportunities. Managers won't have much fun in the process, either, as they will live in constant conflict with creators.

But steward-leaning managers should realize that they might never feel completely comfortable with the inclinations of creators. Differences between stewards and creators are essential; they can't be eliminated. Stewards can know perfectly well on a rational level that they need relentless innovators, people who hunger to create new ideas, try new technologies and do things in different ways. But the challenges in managing someone who never wants to do the same thing twice and who keeps going off on tangents are substantial, and quite different from the more traditional challenges of standardizing work and monitoring compliance. Wise stewards know that they need creators. But they'll probably never like needing creators – their difficult essential partners in business innovation.

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